

Human Multi-Robot Collaboration: Model Predictive Control Based Recommendation System for Reducing Human Cognitive Load

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ABSTRACT

This study presents a human-computer interaction system designed for effective collaboration between a human operator and a team of four robots tasked with mapping unknown terrains on a battlefield. While operating autonomously, robots require human attention and control to avoid obstacles entirely. The human teammate can switch to and take control of any robot, providing timely intervention to help it avoid obstacles. A recommendation system using concepts of modern control theory was developed to support the human operator in critical decision-making, such as determining when to take control of a robot and which to prioritize.



Fig 1: User Interface for Human Operator.

INTRODUCTION

Robots face challenges in critical decision-making during dynamic war scenarios, while humans excel but are prone to fatigue in prolonged operations. Integrating human supervisory control with autonomous robot capabilities for mapping unknown battlefields enhances efficiency and can reduce casualties.

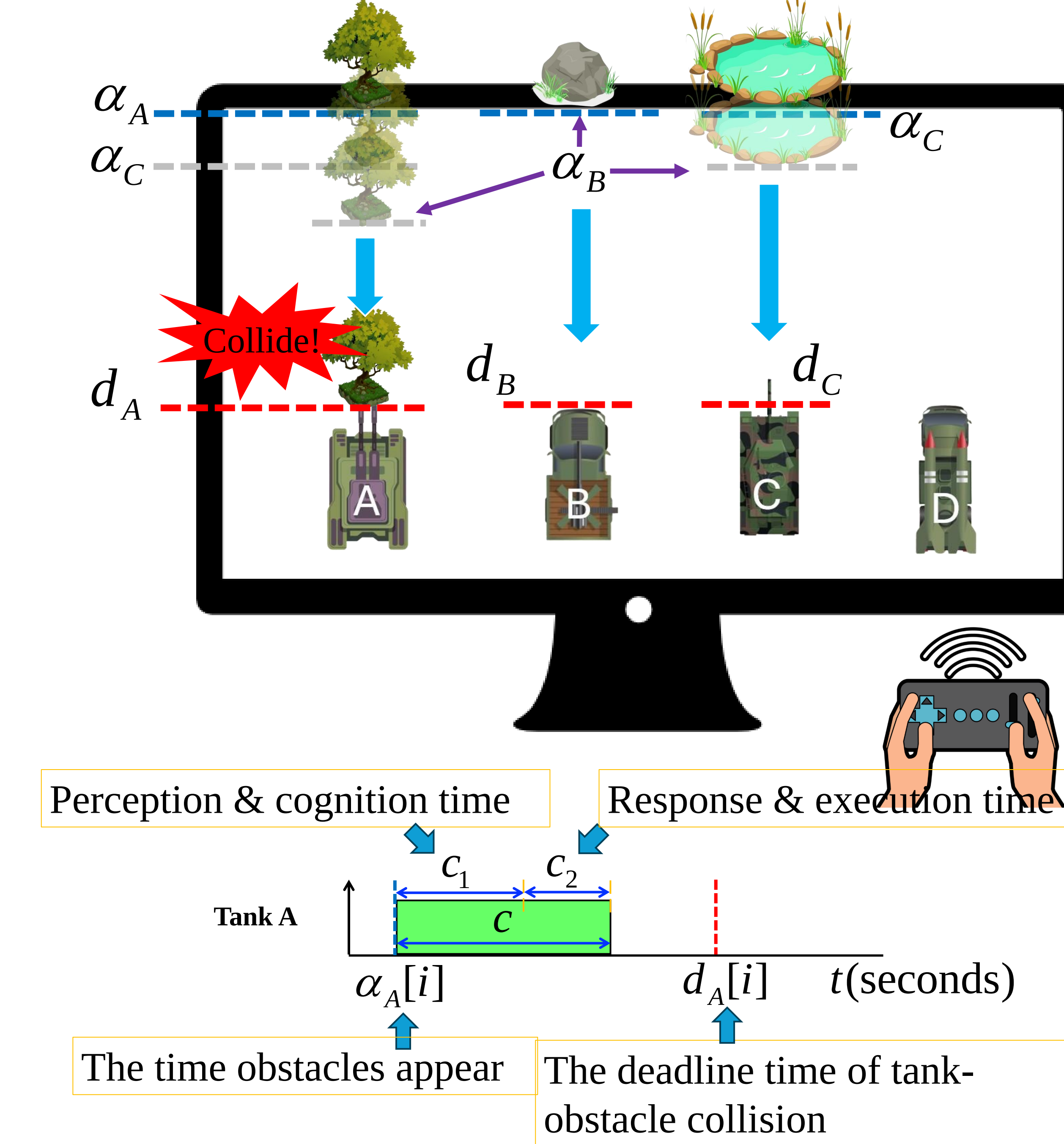
Human operators managing multiple robots face high cognitive load and attention fatigue, reducing mission efficiency. A pilot study with 10 participants (N=10) analyzed human switching times between robots to optimize task prioritization and refine the recommendation system using real-time experimental data for

PROBLEM STATEMENT

The problem involves optimizing human supervisory control of multiple robots for mapping unknown battlefields by minimizing cognitive load and improving task prioritization. Key challenges include limited attention restricting control to one robot at a time, causing potential collisions, and difficulty estimating attention allocation per robot.

METHODOLOGY

Step 1: Establish the task model



Step 2: Optimization Formulation

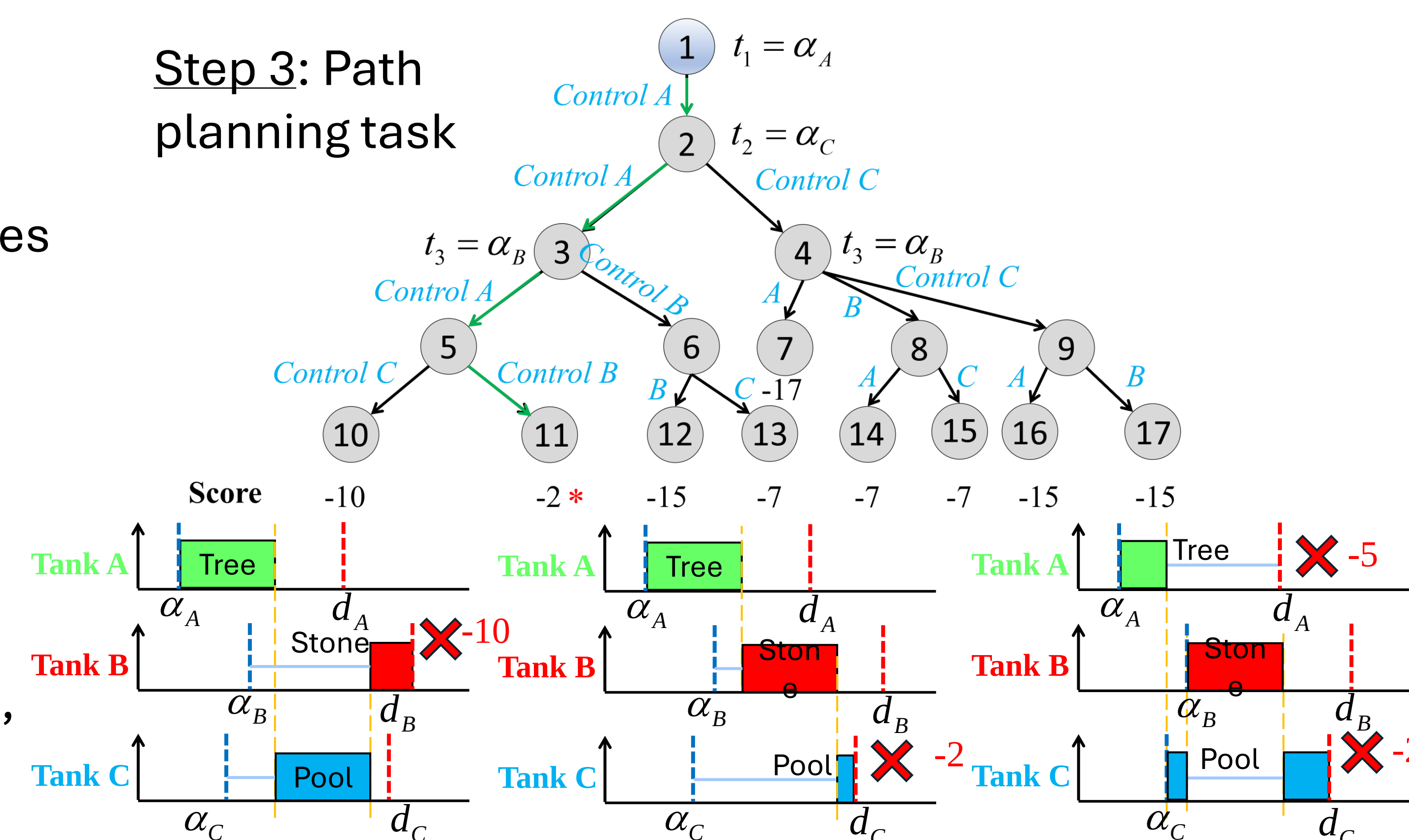
Decide which tank to control at the current time

$$\max_{U(t)} \text{Score} = \sum_{n=1}^N \sum_{i=1}^{I_n} P_n[i]$$

$$s.t. \alpha_A[i] = \alpha_A[i] + (t_{\text{Human}} - t_{\text{Tank}} - t_{\text{Task}}/2)/r$$

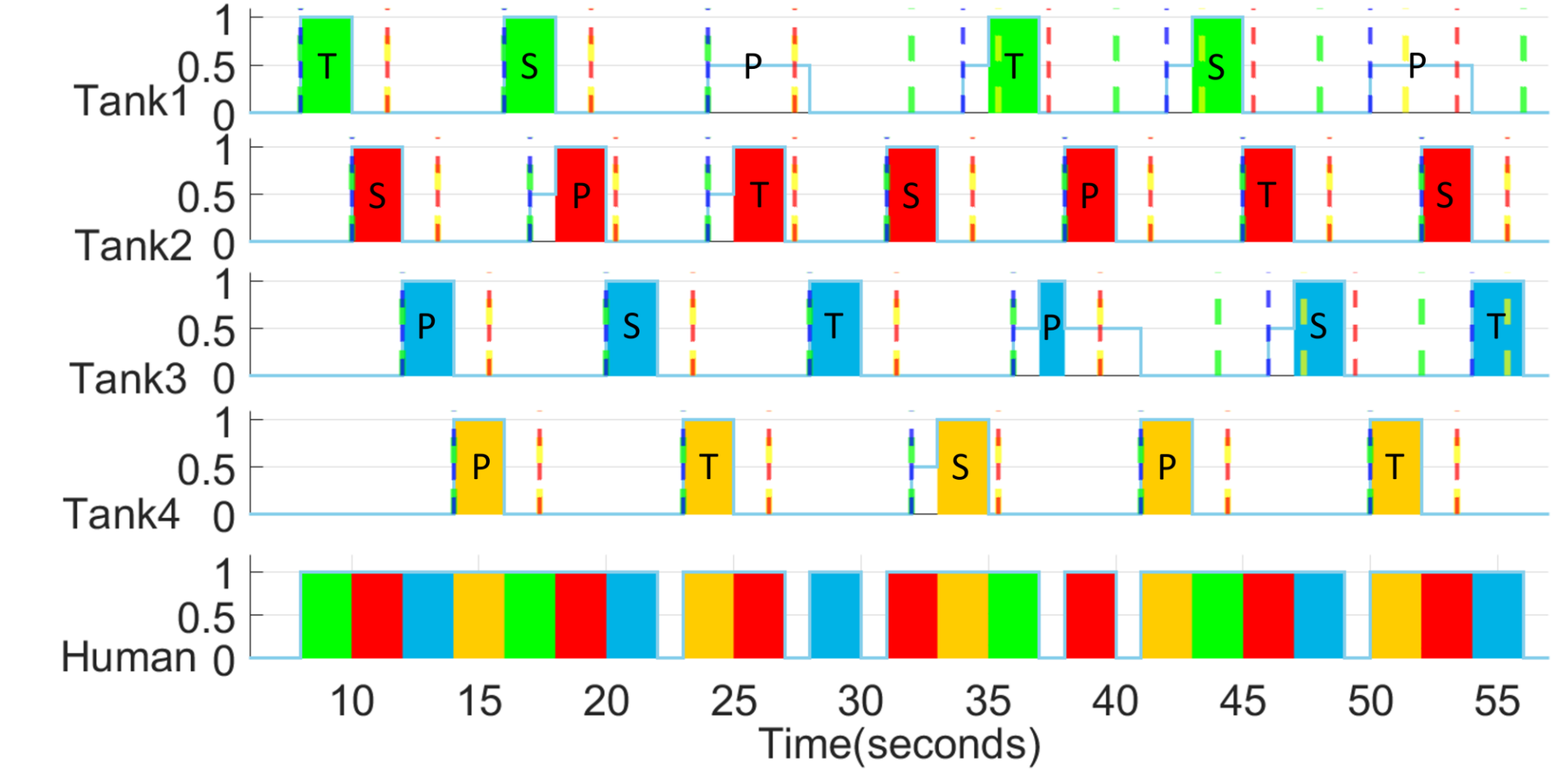
$$P_n[i] = 0, -10, -5, -2, \text{ for all tank } n \text{ and obstacle } i.$$

Step 3: Path planning task



RESULTS

Optimal attention allocation over 4 mins using MPC, with a prediction horizon of 100 seconds.



Switching Time

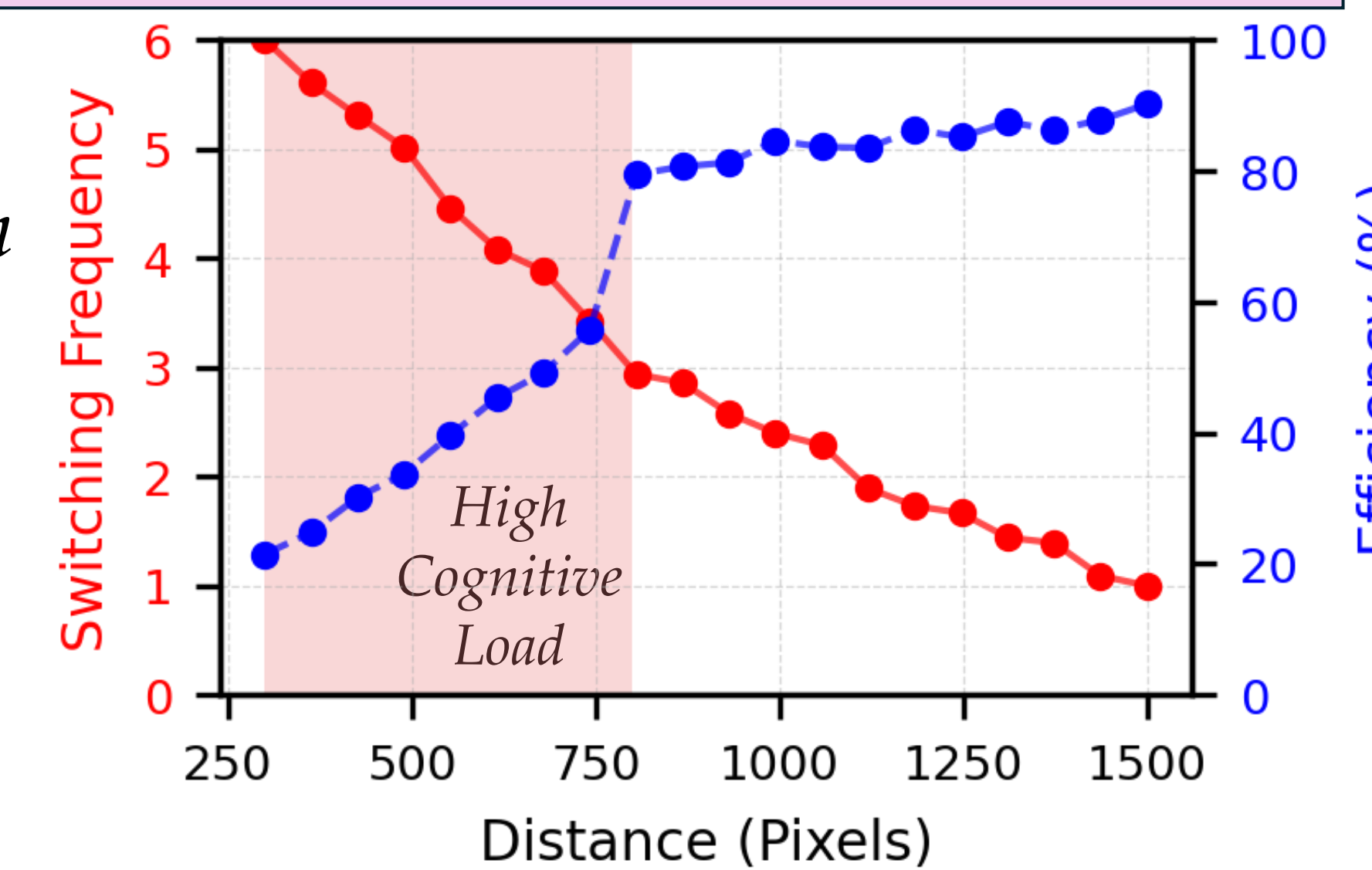
H1: Switching time increases with the number of obstacles, but human efficiency remains unaffected.

User Feed Back Question:
Did recommendation system helped You?

60% Yes

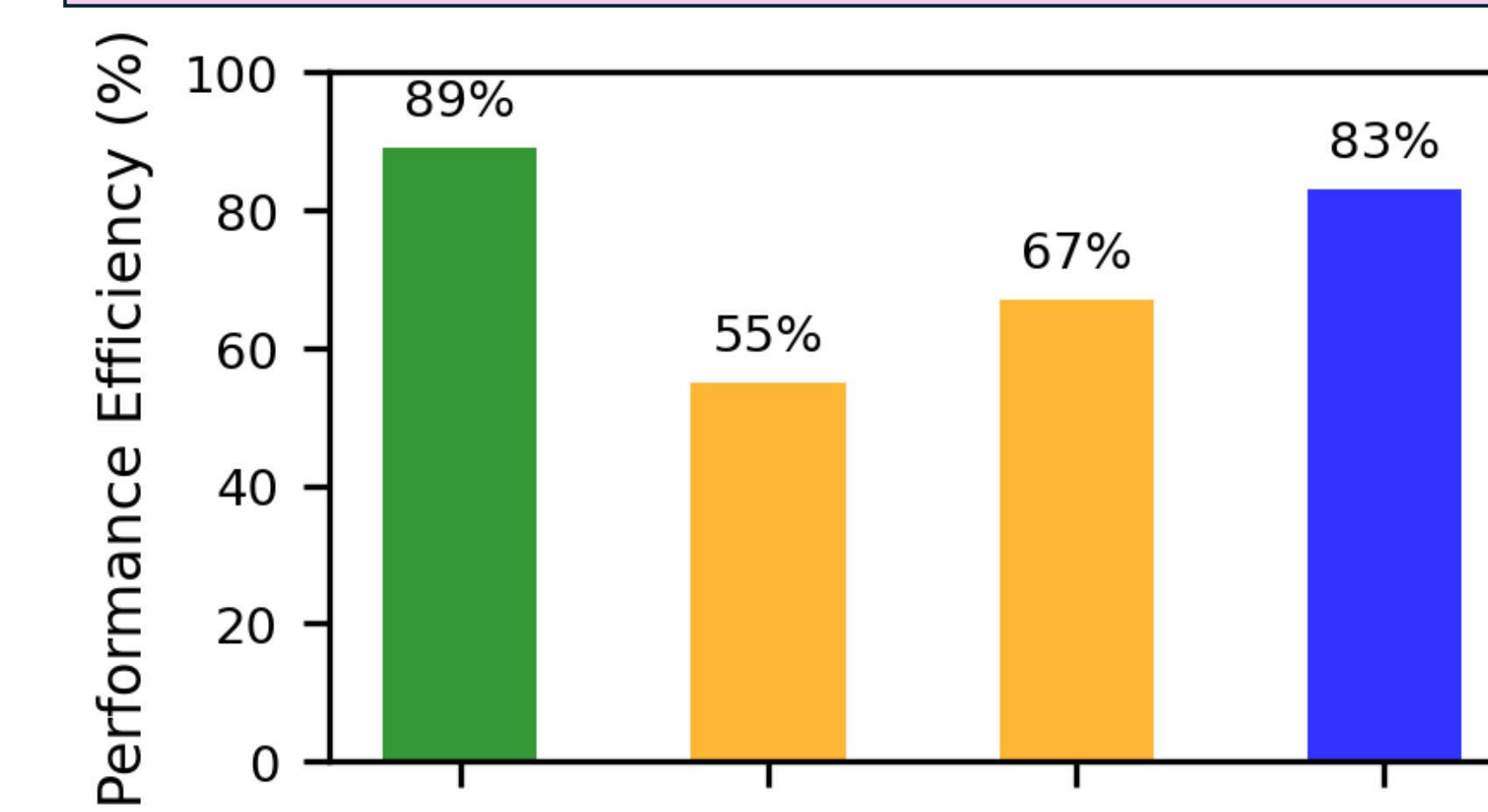
30% No

10% Maybe

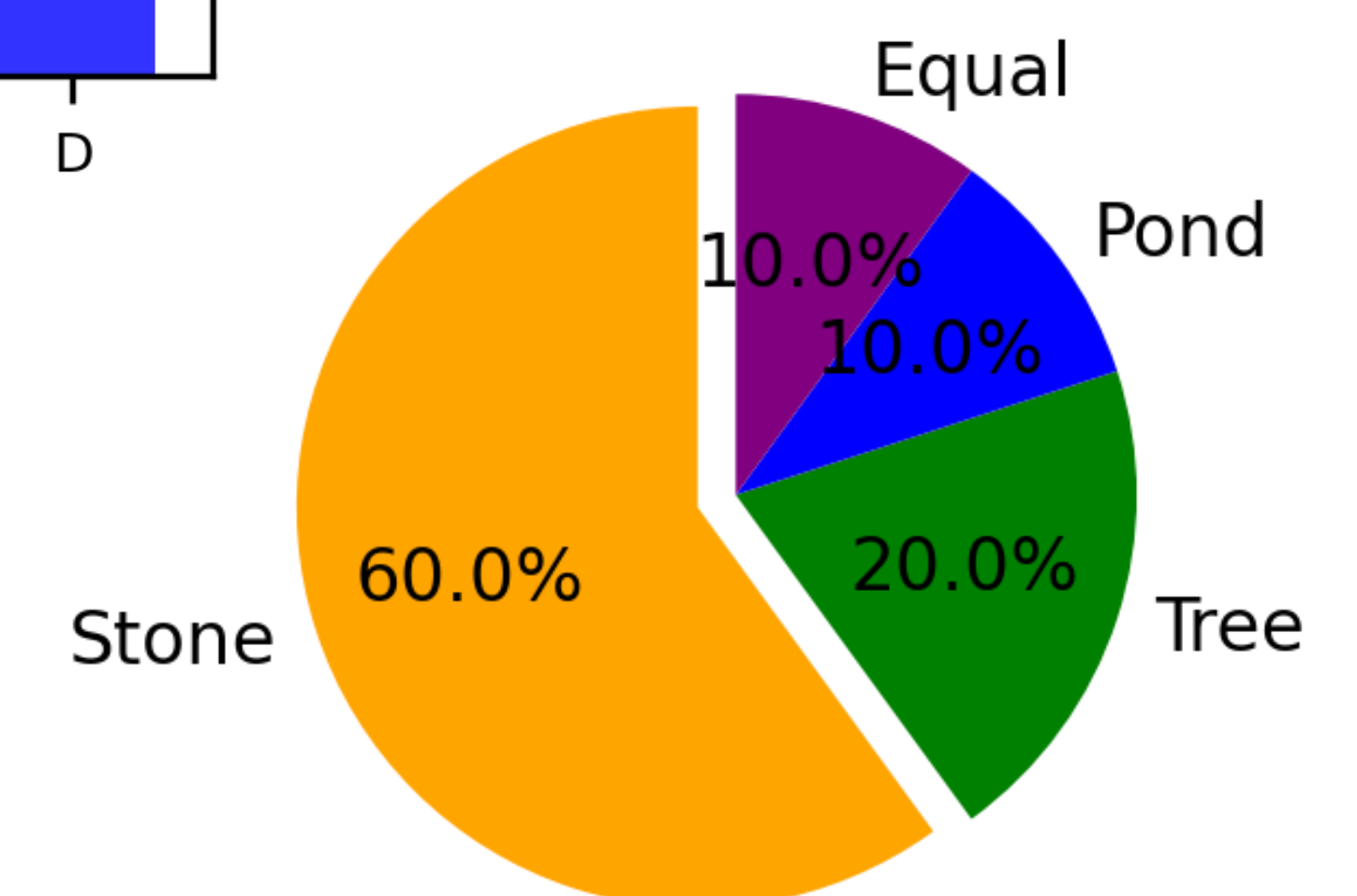


Robot Specific Cognitive Load

H2: Robots located in the middle lanes (B and C) impose a higher cognitive load on humans compared to boundary robots (A and D)



User Feed Back Question: Out of stone, tree and pond which you give more importance?



REFERENCES:

- Cummings, M.L. (2015). Operator Interaction with Centralized Versus Decentralized UAV Architectures. In: Handbook of Unmanned Aerial Vehicles. Springer, Dordrecht.
- N. Yao and F. Zhang, "Optimal Real-time Scheduling of Human Attention for a Human and robot Collaboration System," 2020 American Control Conference (ACC), Denver, CO, USA